Danielle A. Ripsman

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Research Summary: Large-scale and robust optimization for healthcare decisions; learning optimization models and algorithms from data using mixed-integer, Pareto-robust and inverse optimization methods; applications in radiation therapy planning, home care scheduling and the adoption of AI tools in patient-facing care.

■ ACADEMIC APPOINTMENTS

Johns Hopkins University — *2025 – Present*
Malone Postdoctoral Fellow, Malone Center for Engineering in Healthcare
Whiting School of Engineering, Baltimore, MD, USA
Additional Affiliations: *Center for Systems Science and Engineering (CSSE), Hopkins Business of Health Initiative (HBHI)*
Mentors: Dr. Kimia Ghobadi and Dr. Todd McNutt

University of Toronto — *2024 – 2025*
Postdoctoral Fellow in Operations Management & Statistics
Rotman School of Management, Toronto, ON, Canada
Additional Affiliations: *Schulich School of Business, York University; Department of Management, University of Toronto Scarborough*
Mentors: Dr. Andre Cire and Dr. Adam Diamant

■ EDUCATION

University of Waterloo — *2018 – 2024*
Ph.D. in Management Sciences
Management Science & Engineering Department, Waterloo, ON, Canada
Advisor: Dr. Houra Mahmoudzadeh
Thesis: Large-Scale Optimization Approaches for Radiation Therapy

University of Waterloo — *2016 – 2018*
M.A.Sc. in Management Sciences
Management Sciences Department, Waterloo, ON, Canada
Advisor: Dr. Houra Mahmoudzadeh
Thesis: Robust Direct Aperture Optimization Methods for Cardiac Sparing in Left-Sided Breast Cancer Radiation Therapy

University of Toronto — *2010 – 2016*
B.A.Sc. in Industrial Engineering
Mechanical & Industrial Engineering Department, Toronto, ON, Canada
Graduated with Honours and an Engineering Business Minor

■ PUBLICATIONS

Refereed Journal Articles

1. **D. A. Ripsman**, S. H. Hristov, G. Gola, W. Kwong, M. Singh, E. Osei, J. Darko and H. Mahmoudzadeh. "A comparison of voxel sampling approaches for intensity-modulated radiation therapy." *Medical Physics*, 52(12):e70179, 2025. [Link]
2. **D. A. Ripsman**, F. Rahimi, H. Abouee-Mehrizi and H. Mahmoudzadeh. "Light Pareto robust optimization for IMRT treatment planning." *Medical Physics*, 50(5):2637–2648, 2023. [Link]
3. **D. A. Ripsman**, T. G. Purdie, T. C. Y. Chan and H. Mahmoudzadeh. "Robust direct aperture optimization for radiation therapy treatment planning." *INFORMS Journal on Computing*, 34(4):2017–2038, 2022. [Link]
4. **D. A. Ripsman**, D. M. Aleman and K. Ghobadi. "Interactive visual guidance for automated stereotactic radiosurgery treatment planning." *Expert Systems with Applications*, 42(21):8337–8348, 2015. [Link]

Under Review

1. F. Parker*, **D. A. Ripsman***, E. Springer and K. Ghobadi. "Learning warm starts for the simplex method from measured solver cost." Under review at the *International Conference on Learning Representations (ICLR)* 2027. arXiv:2610.20892. [Link] (*Co-first authors.)

Working Papers

1. F. Rahimi, **D. A. Ripsman**, H. Abouee-Mehrizi and H. Mahmoudzadeh. "Superior light Pareto robust optimization with application to radiation therapy." To be submitted to *INFORMS Journal on Optimization*.
2. **D. A. Ripsman**, R. Chiplunkar and K. Ghobadi. "Early reach and workload associations of an ambient AI scribe in community primary care." To be submitted to *JAMA Network Open*.
3. F. Parker, **D. A. Ripsman**, E. Springer and K. Ghobadi. "Pivoting under partial observability: Exact search, learned policies, and degeneracy in simplex pivot selection." To be submitted to *INFORMS Journal on Computing*.
4. **D. A. Ripsman**, A. Cire and A. Diamant. "A decision diagram approach to multi-period home care scheduling under uncertainty." Manuscript in preparation.
5. **D. A. Ripsman**, S. A. Alev and H. Mahmoudzadeh. "A geometric approach to beam angle selection in radiation therapy treatment planning." To be submitted to *European Journal of Operational Research*.
6. **D. A. Ripsman** and H. Mahmoudzadeh. "Closed-form constraint generation for classes of robust polyhedral uncertainty." Manuscript in preparation.

■ PRESENTATIONS

All presented by D. A. Ripsman.

Conference Presentations

1. **D. A. Ripsman**, R. Chiplunkar and K. Ghobadi "The Impact of AI Scribe on Primary Care Workload: A multi-site DiD cohort study", INFORMS Healthcare, Raleigh, NC, July 2026.
2. **D. A. Ripsman** and K. Ghobadi "Generative Inverse Optimization for Healthcare Decision Making", INFORMS Healthcare, Raleigh, NC, July 2026.
3. **D. A. Ripsman**, R. Chiplunkar and K. Ghobadi "The Impact of AI Scribe on Primary Care Workload", CORS Annual Meeting, Kingston, ON, June 2026.
4. **D. A. Ripsman**, A. Cire and A. Diamant. "Multi-Period Home Care Scheduling Under Uncertainty", CORS Annual Meeting, Edmonton, AB, June 2025.
5. **D. A. Ripsman**, "A Geometrical Approach to Beam Angle Optimization", ICS Conference, Toronto, ON, March 2025.
6. **D. A. Ripsman**, F. Rahimi, H. Abouee-Mehrizi & H. Mahmoudzadeh, "Light Pareto Robust Optimization for IMRT Treatment Planning", CORS Annual Meeting, London, ON, June 2024.
7. **D. A. Ripsman**, "Integrating Uncertainty Mitigation into Radiation Therapy Treatment Planning", INFORMS Annual Meeting, Phoenix, AZ, October 2023.
8. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, "A Geometric Approach To Beam Angle Selection In Radiation Therapy Treatment Planning", INFORMS Healthcare, Toronto, ON, July 2023.
9. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, "A Geometric Approach To Beam Angle Selection In Radiation Therapy Treatment Planning", CORS/Optimization Days, HEC Montréal, QC, May 2023.
10. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, "A Geometric Approach To Beam Angle Selection In Radiation Therapy Treatment Planning", INFORMS Annual Meeting, Indianapolis, IN, October 2022.

11. **D. A. Ripsman**, H. Mahmoudzadeh, “Structural Solution Approaches to Robust Optimization with Polyhedral Uncertainty”, CORS/INFORMS International, Vancouver, BC, June 2022.
12. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, “Closed-form Constraint Generation for Classes of Robust Polyhedral Uncertainty”, INFORMS Annual Meeting, Anaheim, CA (Hybrid), October 2021.
13. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, “A Geometric Approach To Beam Angle Selection In Radiation Therapy Treatment Planning”, INFORMS Annual Meeting, Anaheim, CA (Hybrid), October 2021.
14. **D. A. Ripsman**, H. Mahmoudzadeh, “Closed-form constraint generation for robust radiation therapy treatment planning”, CORS Annual Meeting, (Virtual), May 2021.
15. **D. A. Ripsman**, H. Mahmoudzadeh, T. C. Y. Chan, T. G. Purdie, “Robust Direct Aperture Optimization for Radiation Therapy Treatment Planning”, INFORMS Healthcare, Cambridge, MA, July 2019.
16. **D. A. Ripsman**, H. Mahmoudzadeh, T. C. Y. Chan, T. G. Purdie, “Robust Direct Aperture Optimization for Radiation Therapy Treatment Planning”, Canadian Healthcare Optimization Workshop, Saskatoon, SK, May 2019.
17. **D. A. Ripsman**, H. Mahmoudzadeh, T. C. Y. Chan, T. G. Purdie, “Robust Direct Aperture Optimization for Radiation Therapy Treatment Planning”, CORS Annual Conference, Halifax, NS, June 2018.
18. **D. A. Ripsman**, H. Mahmoudzadeh, T. C. Y. Chan, T. G. Purdie, “Robust Mixed Integer Optimization for Radiation Therapy Treatment Planning with Delivery Constraints”, Optimization Days, Montreal, QC, May 2017.
19. **D. A. Ripsman**, K. Ghobadi and D. M. Aleman. “Interactive visual decision support for radiation therapy treatment planning”, Industrial & Systems Engineering Research Conference (ISERC), Montreal, QC, June 2014.
20. **D. A. Ripsman**, K. Ghobadi and D. M. Aleman. “Interactive visual decision support for radiation therapy treatment planning”, Undergraduate Engineering Research Day (UnERD), University of Toronto, ON, August 2013.
    Second place in the Industrial Engineering Oral Presentations category.

Poster Presentations

1. R. Chiplunkar, **D. A. Ripsman** and K. Ghobadi. “The Impact of AI Scribe on Primary Care Workload”, Department of Medicine and Whiting School of Engineering Research Retreat, Baltimore, USA, 2026.
2. R. Chiplunkar, **D. A. Ripsman** and K. Ghobadi. “The Impact of AI Scribe on Primary Care Workload”, Johns Hopkins Research Symposium on Engineering in Healthcare, Baltimore, USA, 2025.
3. **D. A. Ripsman**, K. Ghobadi and D. M. Aleman. “Graphical Interface for Interactive Parameter Selection in Radiation Therapy Treatment Planning”, MIE Research Symposium, University of Toronto, Toronto, Canada, June 2013.

■ FUNDING

Fellowships & Scholarships

- *2025 – Present* ▪ Malone Postdoctoral Fellowship, Malone Center for Engineering in Healthcare, Johns Hopkins University
- *2019 – 2022* ▪ NSERC Canada Graduate Scholarship–Doctoral (CGS-D) ($105,000; $35,000 for 3 years)
- *2018 – 2022* ▪ Engineering Excellence PhD Fellowship ($120,000; $30,000 for 4 years)
- *2018* ▪ Provost’s Doctoral Entrance Award for Women ($5,000)
- *2017* ▪ University of Waterloo Graduate Scholarship ($750)

2013 ■ NSERC Undergraduate Student Research Award ($4,500)

2010 – 2011 ■ John Hirschorn Memorial Scholarship ($20,000; $10,000 for 2 years)

2010 ■ University of Toronto Scholars Scholarship ($5,000)

■ AWARDS & HONOURS

Teaching Awards

2020, 2022 ■ Sandford Fleming Foundation Teaching Assistant Award, for the R laboratory of the introductory statistics course, University of Waterloo

Competitions & Honours

2024 ■ **Finalist** (Health Care Operations Research SIG and Open categories), CORS Student Paper Competition, London, ON, for "Superior and light Pareto robust optimization"

2019 ■ **Winner**, CORS Health Care Operational Research SIG Student Presentation Competition, Saskatoon, SK

2018 ■ **Finalist**, Three Minute Thesis (3MT), University of Waterloo (first place in the departmental heat)

2014 ■ Third place, Simulation Competition, IIE National Student Conference, Toronto, ON

2013 ■ Second place, Industrial Engineering Oral Presentations, Undergraduate Engineering Research Day, University of Toronto, ON

2013 ■ Third place, Operations Research Case Competition, Industrial Engineering Competitions Day, University of Toronto, ON

2012 ■ First place, Student Design Competition, American Society of Mechanical Engineers, University of Toronto, ON

■ GRANT WRITING EXPERIENCE

NIH R01 (NHLBI)
2026 – 2030

SCH: Intelligent Decision-Support System for ICU Respiratory Management
R01HL192731, NSF–NIH Smart and Connected Health program. PIs: Dr. James Fackler, Dr. Kimia Ghobadi and Dr. Chien-Ming Huang.
Contributed to proposal development and writing.

■ TEACHING EXPERIENCE

Course Instructor
May – Aug 2023

MSCI 331: Introduction to Optimization, University of Waterloo
Sole instructor of an evening section; 112 upper-year undergraduates

- **Student Course Perceptions: 4.29/5 average across the four instructor characteristics**, above the Faculty of Engineering average of 3.98 (item range 4.04–4.55; 29 of 112 responding)
- Designed and administered online auto-graded assignments, written quizzes, midterms and final examinations; supervised two teaching assistants
- Held full administrative responsibility for the section, including announcements, grade entry and documentation

Guest Lecturer
Fall 2026

EN.560.450/650: Introduction to Operations Research
Johns Hopkins University, in Dr. K. Ghobadi's course

Teaching Assistant — **Dept. of Management Science & Engineering, University of Waterloo**
Twelve terms across eight courses, 2017–2023

- *2019 – 2023* — *MSCI 251: Probability & Statistics for Engineers I*. Designed and taught the applied R laboratories, and rebuilt them for remote delivery during the pandemic (5 terms; Dr. H. Mahmoudzadeh)
- *2023* — *MSCI 734: Network Models*. Assignment and project grading (Dr. F. Gzara)
- *2022* — *MSCI 435: Advanced Optimization Techniques*. Tutorials, grading and student support (Dr. S. Elhedhli)
- *2022* — *MSCI 703: Applied Optimization*. Grading and solution manuals (Dr. S. Elhedhli)
- *2019* — *MSCI 401/402: Design Project*. Course logistics and discussion sessions (Dr. A. Hurst)
- *2018* — *Work Report Marking*. Assessment and written feedback on work-term reports (D. Chan)
- *2017* — *MSCI 436: Decision Support Systems*. Built and taught weekly Visual Basic tutorials, office hours, grading; 42 students (Dr. M. Karimi)
- *2017* — *MSCI 331: Introduction to Optimization*. Tutorials, office hours, grading and posted solutions; 41 students (Dr. A. Hiassat)

Courses Prepared to Teach

- *Taught:* introduction to optimization, and linear and integer programming (sole instructor); probability and statistics for engineers (5 terms of R laboratories)
- *Foundations:* introductory programming (C++, Java, VB, MATLAB, Python); applied linear algebra
- *Advanced:* optimization under uncertainty (robust and stochastic); large-scale optimization; simulation
- *Applied:* healthcare systems engineering and analytics; decision support systems

■ RESEARCH MENTORSHIP

Undergraduate Researchers — **Dept. of Management Science & Engineering, University of Waterloo**
Six undergraduate co-op students and research assistants supervised, 2018–2023

- *2023* — **Gatik Gola** (co-op): motivated the project, recommended methods and ran weekly meetings for a CT voxel sampling study; *co-author, Medical Physics 2025*
- *2023* — **Karina Wilk** (co-op): onboarding and weekly planning for the hospital optimization pipeline begun during the Grand River Hospital internship
- *2021 – 2022* — **William Kwong** (research assistant): proposed and oversaw an interface for running sampling protocols for fluence map optimization; *co-author, Medical Physics 2025*
- *2021* — **Tiantian Li** (research assistant): exploratory project adapting open-source radiation therapy tools (CERR, matRad) to optimization software
- *2019* — **Gogilan Selvarajah** (research assistant): pair programming and project direction for a sliding-window IMRT visualization
- *2018* — **Jessie Yeung** (research assistant): background research and data acquisition for an IMRT sliding-window modelling project

RESEARCH & CLINICAL EXPERIENCE

Graduate Researcher — *Sept 2016 – Aug 2024*
Dept. of Management Science & Engineering, University of Waterloo
Supervised by Dr. Houra Mahmoudzadeh.
- Built optimization models for radiation therapy treatment planning, tailored to breathing motion uncertainty in breast cancer treatment
- Prototyped and tested models against large clinical datasets using a custom C++/CPLEX code base
- Implemented structured visualizations of plan quality in MATLAB

Clinical Research Intern — *Oct – Dec 2018*
Medical Physics Department, Grand River Hospital, Kitchener, ON
Supervised by Dr. Ernest Osei.
- Built the I/O bridge between Varian Eclipse and open-source CERR modules for importing clinical DICOM plans
- Wrote custom collimator rotation code and validated the pipeline on patient plans

Thesis Student — *Sept 2015 – Apr 2016*
Toronto Intelligent Decision Engineering Laboratory, University of Toronto
Supervised by Dr. J. Christopher Beck.
- Developed A*-based batch scheduling algorithms, with C++ implementations of Dijkstra and A*

Capstone Student — *Sept 2015 – Apr 2016*
Addiction Medicine Services, Centre for Addiction and Mental Health, Toronto, ON
Supervised by Dr. Michael Carter.
- Streamlined clinic scheduling and resource allocation, and built a simulation to validate the recommended interventions

Undergraduate Researcher — *Fall 2012 – May 2014*
Medical Operations Research Laboratory, University of Toronto
Supervised by Dr. Dionne Aleman.
- Took two interactive radiosurgery planning prototypes from concept to completion in self-taught MATLAB

Case Study Author — *Fall 2014*
Industrial Engineering Department, University of Toronto
Case adopted as a teaching aid in Cases in Operational Research (MIE367).
- Wrote a multi-criteria decision-making case around a restaurant venture, with realistic data distributions and a financial planning simulation

PROFESSIONAL SERVICE

Journal Refereeing
Referee for *Computers & Operations Research*, *European Journal of Operational Research*, *International Transactions in Operational Research*, *Medical Physics*, *Operations Research Forum* and *Quantitative Imaging in Medicine and Surgery*

Conference Service
- Session Chair, INFORMS Computing Society (ICS) Conference, Toronto, ON, 2025; also a conference volunteer
- Session Chair, CORS/INFORMS International Conference, Vancouver, BC, 2022
- Designed the virtual conference venue (Gather Town) for the CORS Annual Conference, 2021, with Dr. Kiefer Burgess

Memberships
Institute for Operations Research and the Management Sciences (INFORMS), including its Health Applications Society (HAS); Canadian Operational Research Society (CORS)

LEADERSHIP & OUTREACH

National Student Councillor — *2024 – 2025*
Canadian Operational Research Society (CORS)
Elected to represent the national student body.
- Led the post-pandemic revival of student chapters across Canada

Invited Session Leader — *May 2024*
SPARCS Workshop (CEMC), University of Waterloo
Week-long computer science workshop for Grade 9 and 10 students of gender identities underrepresented in computing.
- Adapted the TORCH format into a two-hour introduction to operations research for 40 students: ten teams competed on two optimization puzzles, followed by a talk linking their strategies to real-world problems

President — *Sept 2017 – 2020*
The Operations Research Challenge (TORCH), University of Waterloo
Founded the Waterloo-region edition of the annual high school competition.
- Led logistics for the 2018, 2019 and *2020*[*] competitions: student and volunteer recruitment, funding, testing and scheduling
- Wrote competition problems and validators, and coordinated the automated grading software, with volunteers from Waterloo, Concordia and Toronto

Student Chapter President — *Sept 2017 – Sept 2018*
Canadian Operational Research Society (CORS), Waterloo, ON
Ran the local student chapter's speaker and social programme.
- Coordinated speakers and events, onboarding, communications and website oversight
- Grew active membership from fewer than 10 to nearly 30

President — *Sept 2016 – Aug 2017*
Management Sciences Student Association, University of Waterloo
Elected departmental representative for graduate students.
- Budgeted and ran invited speaker and social events; represented students at monthly departmental and cross-departmental meetings

Executive Member — *Fall 2014 – Winter 2016*
The Operations Research Challenge (TORCH), University of Toronto
Co-Chief Creative Officer (2016); Assistant Editor-in-Chief (2015).
- Oversaw creation of the 2016 contest, edited the 2015 problem set, and wrote a scheduling problem with an automated solver for grading

INDUSTRY EXPERIENCE

Rational Software Developer — *May 2014 – Aug 2015*
IBM Canada Software Lab, Markham, ON
Eclipse-based enterprise software development, testing and debugging (TPF Toolkit, RDi); supervised by Rob Cecco.

Engineering Intern — *Summer 2012*
Rotork Controls Canada, Mississauga, ON
Engineering drawings, customer documentation, and a Visual Basic database for intracompany drawing retrieval.

TECHNICAL SKILLS

Languages & Tools — Python, C++, Java, MATLAB, R, LaTeX

Optimization — CPLEX, Gurobi, HiGHS, AMPL, Excel Solver

Clinical / Domain — Varian Eclipse, CERR, matRad, DICOM-RT

[*]Cancelled due to the pandemic